

MAR657: Visual Analytics

Fall 2025

Instructor information

Instructor

Eunhee (Emily) Ko

Email

eko2@pace.edu

Classroom location & hours

1 Pace Plaza W623, 1:20 PM-4:20 PM
(Wednesdays)

Office hours

Wednesdays: 12:00 PM - 1:00 PM or by appointment (Please make an appointment, in-person or virtual)

Zoom info:

<https://pace.zoom.us/j/9968104723>

Meeting ID: 996 810 4723

Password: 249255

General information

Description

Visual analytics will introduce you to broad classes of techniques and tools for analyzing and visualizing data at scale. In this course, students will work on solving complex problems in data science using exploratory data visualization and analysis in combination. It will employ visualization software such as Tableau or R. The pedagogical philosophy of this course is built around the principle of learning by doing. Each visualization technique will be discussed at a high level. Next, the technique will be implemented on a large dataset in class in an interactive manner. This will be followed by a hands-on assignment in which you will use the same technique to tackle a different problem. Finally, you will have the opportunity to demonstrate your proficiency by completing a final predictive analysis project and a presentation.

Learning outcomes

This course is designed to develop skills in getting, exploring, manipulating, analyzing, and presenting business data using data visualizations. Specifically, this course will teach the student how to

- visually explore, manipulate and analyze data
- communicate analysis results using effective visualizations
- develop skills in converting data and information into insights and decisions
- tell compelling stories with data
- use data visualization software (e.g., R, Tableau)

Text, readings, and software resources

- Instructional materials
 - Storytelling with Data, Cole Nussbaumer-Knaflic, 2015. ISBN: 978-1119002253
 - Visual Data Storytelling with Tableau, Lindy Ryan, 2018. ISBN: 978-0134712833
 - Some parts of the lecture slides are instructor-created resources on the reference: Yong Woo Kim, Do it! R Data Analysis, EasysPublishing, 2017
- R resources

- i) Lynda.com (e.g., 'Learning R with Barton Poulson'; Pace University students have free access to Lynda.com using NetID and password. Type R in the search bar after logging in. It shows numerous tutoring videos for R)
- ii) <https://cran.r-project.org/manuals.html>
- R or RStudio download
<https://www.r-project.org/>
<https://www.rstudio.com/products/rstudio/download/>
- Tableau desktop
Students should download and install the free version of Tableau for class use here:
<https://www.tableau.com/academic/students>

Grading

Grading allocation

- Attendance and participation (50 points)
- 10 individual assignments (40 points each = 400 points)
- Mid-term and final quizzes (100 points each = 200 points)
- Final group project deliverable and presentation (150 points each = 300 points)
- Peer evaluation (50 points)

Grading guidelines

A: 930 - 1000 points
A-: 900 - 929 points
B+: 850 - 899 points
B: 780 - 849 points
B-: 750 - 779 points
C+: 700 - 749 points
C: 550 - 699 points
C- : 500 - 549 points
D: 250 - 499 points
F: 0 - 249 points

Grading policies

- **Late assignments:** Late assignments are not usually accepted, but with tangible evidence, the following late assignment penalty may be applied:
 - A 5% point deduction from the total points for an assignment (e.g., 38/40) submitted within 24 hours of the deadline
 - A 15% point deduction from the total points for an assignment (e.g., 34/40) submitted within 72 hours of the deadline.
 - A 25% point deduction from the total points for an assignment (e.g., 30/40) submitted within 168 hours (1 week) of the deadline.
 - A 50% point deduction from the total points for an assignment (e.g., 20/40) submitted later than 1 week after the deadline.
- **Makeup exam:** No makeup exam is accepted.

Class preparation, assignments and communication

Class preparation and management

Be sure that you properly prepare for each class. For example, there are a couple of weeks when you will need to read articles before the session. You may need to have a laptop for lectures to do a class

exercise or discuss an assignment during the lecture. For example, after I show you certain exercise, how to visualize cluster analysis results using R, I may ask you to do a couple of related exercises using R. In this case, you may need to have a laptop with you and open R Studio during classes. This method not only facilitates your learning by doing, but it also gives you the opportunity to ask questions about the lecture and assignments.

Assignments and class activities

- Class attendance and participation
 - Your participations will be evaluated based on class attendance and class participation.
 - Unexcused absences in excess of **two** scheduled lectures (i.e., **three-** or **four-**times unexcused absences) will result in **5%** grade reduction from the total score. Unexcused absences in excess of **four** scheduled lectures (i.e., **five-** or **six-**times unexcused absences) will result in **10%** grade reduction from the total score. Unexcused absences in excess of **six** scheduled lectures (i.e., more than **six-**times unexcused absences) will result in **30%** grade reduction from the total score. Students with unusual circumstances should advise the instructor of their situation immediately.
 - Class meeting attendance check routine
 - Sign your name on the attendance sheet as you enter the classroom.
 - The attendance sheet will be removed 15 minutes after the class starts, and you will no longer be able to sign in after that time.
 - If you arrive more than 15 minutes late, please see me during the break to address your attendance.
 - Late sign-ins will be marked as “tardy.”
 - **Three unexcused tardies equals one absence.**
 - Students are expected to be present for the entire class period and should not leave early without prior approval from the instructor.
 - Leaving class early without permission may result in a penalty, such as a deduction from participation points.
 - For three-hour classes, there will be one or two 10-minute bathroom breaks.
 - In general, students are expected to demonstrate respect and take responsibilities for managing their own behavior.
- 10 individual visualization assignments
 - The deliverable containing all the data analyses and interpretations (3 - 5 pages) is found in the file named ***YourName_Individual Visual Analytics Assignment XX.pdf***.
 - Students should use the assignment deliverable form that is part of the assignment document.
 - As with all of the Visual Analytics Assignments, you can work with others, but your submission must completely be your own work. Do not send an electronic or paper copy of your work to anyone. Do not receive electronic or paper copies from anyone via email or online websites. Doing this is considered cheating, even if the intention is not to.
 - The assignments will be compared for plagiarism with at least two different methods. Students suspected of sharing work will receive a zero and possibly a zero in the course.
 - Assignment grading rubric:
 - N: Nothing turned in (0 points)
 - B: Below expectations - something turned in, but of poor quality (25 points)
 - M: Meets the expectations of good work in this class (33 points)
 - E: Exceeds expectations - something that is truly notable (40 points)Note: What makes the assignment excellent? Completeness, correctness, organization, and aesthetics
- Mid-term and final quizzes
 - The test consists of multiple-choice questions, true or false, and short answer questions.

- A study guide will be distributed prior to the test.
- The mid-term and final tests are closed book, but you are allowed one one-sided page (up to 8.5 by 11 inches) of notes for each test.
- Final group project deliverable and presentation
 - Students can choose their own data.
 - Students can choose the analysis method, but the analysis should contain both **exploratory** and **two or three visual analytics methods** that are learned from the course.
 - 20-min or 25-min presentation per group (the allotted time can change depending on the number of groups).

All students must present since the grade for a presentation varies for each student.

Brightspace

It is critically important to check Brightspace frequently. I will post class materials and important course information such as announcements, office hours, and due dates on Brightspace. Please check Brightspace frequently for updates.

Communication

I will use email to communicate with you between class meetings. You are responsible for regularly checking your Pace email account and Brightspace. If you need to communicate with me outside of class or office hours, **email** is the preferred means of communication. I will try to respond as soon as possible, but please expect **12-24 hours** to get a reply. Before sending an email, be certain that your concerns cannot otherwise be resolved (e.g., by consulting the syllabus, course materials, or assignment details).

Course outline

Module	Date		Topic	Evaluation
1	9/3/2025	Wed	Course introduction and data visualization + Final project introductory workshop	Attendance and participation
2	9/10/2025	Wed	Data visualization: Foundations and theories	Assignment 1
3	9/17/2025	Wed	Introduction of R	Assignment 2 Group forming
4	9/24/2025	Wed	Exploratory data analysis (EDA) and visualization	Assignment 3
5	10/1/2025	Wed	Data manipulation, filtering, and visualizations	Assignment 4
6	10/8/2025	Wed	Presenting results from hypothesis testing and data analysis I + Final project proposal meeting I	Final project proposal
7	10/15/2025	Wed	Presenting results from hypothesis testing and data analysis II+ Final project proposal meeting II	Assignment 5
8	10/22/2025	Wed	Maps and interactive graphs Final project proposal meeting III	Assignments 6 + 7
9	10/29/2025	Wed	Midterm quiz: Take-home	Midterm quiz
10	11/5/2025	Wed	Tableau I	Assignment 8

11	11/12/2025	Wed	Tableau II + Tableau III Aesthetic and crafting the story	Assignments 9 + 10
12	11/19/2025	Wed	Final group project presentation I	Presentation
	11/26/2025	Wed	Thanksgiving break – No class meeting	
13	12/3/2025	Wed	Final group project presentation II	Presentation
14	12/10/2025	Wed	Final quiz: Take -home	Final quiz
15	12/ 17/ 2025	Wed	Final project deliverable due	Final project deliverables

Note 1. The instructor reserves the right to make changes to the course syllabus over the course of the quarter.

Note 2. All assignments and required submissions should be uploaded to Brightspace by the time that is specified above.

University Policies and Resources

Academic Integrity

Students in this course are required to adhere to Pace University's Academic Integrity Code. The Academic Integrity Code supports honesty and ethical conduct in the educational process. It educates students about what constitutes academic misconduct, helps to deter cheating and plagiarism, and provides a procedure for handling cases of academic misconduct. Students are expected to be familiar with the Code, which can be found under "University Policies" in the

[Student Handbook](#). Individual schools and programs may have additional standards of academic integrity. Students are responsible for familiarizing themselves with the policies of the schools, programs, and courses in which they are enrolled.

Instructions for faculty, a paper about best practices, and the form for reporting integrity policy violations are posted on the [Provost's webpage under policies and forms](#). Reporting forms should be sent to the Chairs of the Academic Conduct Committee on each campus.

Learning Centers

[The Learning Center](#) uses an array of programs and a holistic approach to assist students with academic skills and content knowledge. We are dedicated to developing independent learners by creating purposeful interactions with trained, well-qualified peer and professional staff.

Procedure for Students Who Wish to Obtain Reasonable Accommodations for a Course:

The University's commitment to equal educational opportunities for students with disabilities includes providing reasonable accommodations for the needs of students with disabilities. To request a reasonable accommodation for a qualified disability a student with a disability must self-identify and register with Student Accessibility Services for his or her campus. No one, including faculty, is authorized to evaluate the need for or grant a request for an accommodation except Student Accessibility Services. Moreover, no one, including faculty, is authorized to contact Student Accessibility Services on behalf of a student. For further information, please see [Resources for Students with Disabilities page](#).

Technological Resources:

- List of all [Pace Information Technology Services](#).

- For assistance with a technological concern (Blackboard, Internet, Computer, etc.), contact the Pace Helpdesk at 914-773-3648 or create a [help desk ticket](#).
- Visit the [Learning Remotely website](#)

Appropriate Use Policy for Information Technology:

Pace endorses the following statement on software and intellectual rights distributed by EDUCAUSE, the non-profit consortium of colleges and universities, committed to the use and management of information technology in higher education. The statement reads:

Respect for intellectual labor and creativity is vital to academic discourse and enterprise. This principle applies to work of all authors and publishers in all media. It encompasses respect for the right to acknowledgment, right to privacy and right to determine the form, manner and terms of publication and distribution.

Because electronic information is volatile and easily reproduced, respect for the work and personal expression of others is especially critical in computer environments. Violations of authorial integrity, including plagiarism, invasion of privacy, unauthorized access and trade secret and copyright violations, may be grounds for sanctions against members of the academic community.

[Pace's appropriate use policy](#) applies to recordings of classroom instruction and digital artifacts created by faculty and students.

Sex-Based Misconduct Policy and Procedure:

Pace University is committed to providing a safe environment for every member of its community and to ensuring that no student, faculty or staff member is excluded from participation in or denied the benefits of any University program or activity on the basis of sex. Accordingly, the University prohibits the following forms of Sex-Based Misconduct: sexual assault, sexual harassment, gender-based harassment, dating violence, domestic violence, sexual exploitation and stalking.

Instructors are a **non-confidential** resource and have an obligation to report any information about sexual assault with the Executive Director of Institutional Equity and Title IX Coordinator (Bernard Dufresne, bdufresne@pace.edu, 163 Williams Street, Room 1017, 212-346-1310). The Title IX/Affirmative Action Office is responsible for investigating violations of the sexual misconduct policy. For more information about the Pace University sexual misconduct policy, see the [Sex-Based Misconduct Policy and Procedure \(PDF\)](#).

Members of the University community who believe that they have been subjected to Sex-Based Misconduct are encouraged to report such incidents to the University and, where applicable, to local law enforcement. **Confidential** resources include the **University Counseling Centers, Offices of Sexual and Interpersonal Wellness** and **University Healthcare**. Contact information for those offices may be found in the self-care section below.

Self-Care:

Your academic success in this course and throughout your college career depends heavily on your personal health and well-being. Stress is a common part of the college experience, and it often can be compounded by unexpected life changes outside the classroom. The Pace Community strongly encourages you to take care of yourself throughout the term, before the demands of midterms and finals reach their peak. Please feel free to talk with me about any difficulty you may be having that may impact your performance in this course as soon as it occurs and before it becomes unmanageable. Please know there are a number of other support services on campus that stand ready to assist you. I strongly encourage you to contact them when needed.

Just In Case App:

The Counseling Center's *Just In Case* App supplies potentially life-saving mental health information to Pace University students, staff, and faculty. This smart phone App puts vital information and support

options at your fingertips. Scan and open the App today, just in case you or a friend needs help. Download the Counseling Center [*Just In Case App*](#) or go to "Counseling Center: Just In Case" on the MyPace Mobile App.

Department	Pleasantville	New York City
Counseling Center	914-773-3710	212-346-1526
Dean for Students Office	914-773-3351	212-346-1306
Health Care Unit	914-773-3760	212-346-1600
Residential Life	914-923-2791	212-346-1295
Student Development and Campus Activities	914-773-3861	212-346-1590
Office of Multicultural Affairs & Diversity Programs	914-773-3775	212-346-1563
Sexual Assault Prevention & Education	914-597-8783	212-346-1931
Academic Advisement		
Advising Center for Exploring Majors	914-773-3847	212-346-1798
College of Health Professions	914-773-3961	914-773-3552
Dyson College	914-773-3781	212-346-1518
International Student / Scholars	914-773-3425	212-346-1368
Lubin School of Business	914-773-3531	212-618-6550
Pforzheimer Honors College	914-773-3941	212-346-1697
Seidenberg School	914-773-3254	212-346-1864
Study Abroad	914-773-3447	212-346-1368